DATA ENGINEERING • HEALTHCARE ANALYTICS

Medallion-Based Healthcare Analytics & Prediction Platform

Real-Time Patient Vitals Monitoring • Early Deterioration Detection

Apache Kafka | PySpark | Delta Lake | Databricks | Azure Data Factory | ADLS

ARCHITECTURE
Medallion (Bronze → Silver → Gold)

DOMAIN
Healthcare / Clinical Operations

CLASSIFICATION
Confidential — Client Delivery

DOCUMENT TYPE
Technical Project Report

Prepared by: Data Engineering Team

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Executive Summary

This report presents the design and implementation plan for a **Medallion-Based Healthcare Analytics and Prediction Platform** — a scalable, real-time data engineering system built to continuously monitor patient vital signs, detect early signs of clinical deterioration, and deliver actionable alerts to medical teams.

The platform leverages a three-layer Medallion Architecture (Bronze → Silver → Gold) deployed on Azure cloud infrastructure, with PySpark-powered transformations and a machine learning layer for risk scoring. The end objective is to reduce delayed interventions and improve patient outcomes through data-driven clinical intelligence.

> *"Built a system to analyze patient vitals in real time and predict early signs of deterioration, enabling faster medical intervention."*

TECHNOLOGY STACK

Component	Technology	Role
Streaming	**Apache Kafka**	Real-time vitals ingestion from bedside devices
Processing	**PySpark / Databricks**	Distributed data transformation across all layers
Storage	**Delta Lake / ADLS**	ACID-compliant lakehouse storage
Orchestration	**Azure Data Factory**	Pipeline scheduling and monitoring
Analytics	**SQL / Gold Layer**	Business KPIs and clinical reporting
ML / Scoring	**Databricks ML**	Patient deterioration risk scoring

1 Problem Statement

> **Critical Gap:** In hospitals, patient conditions can deteriorate rapidly due to sudden changes in vital signs — heart rate, oxygen saturation (SpO2), and blood pressure. Traditional monitoring systems often **fail to detect early warning signs**, leading to delayed medical interventions and increased patient risk.

There is currently no centralized, real-time system that can continuously monitor patient vitals at scale, correlate multiple physiological signals simultaneously, and surface early deterioration patterns before they become life-threatening emergencies. This gap increases the burden on clinical staff who must manually review alerts from multiple disconnected systems.

The platform is designed to address three core failures in current systems:

- **Reactive monitoring:** Alerts are triggered only after critical thresholds are breached, leaving no window for preventive action.
- **Data silos:** Vital signs, patient records, and ICU capacity data exist in separate systems with no unified view.
- **No predictive capability:** Existing systems lack machine learning models capable of scoring deterioration risk in real time.

2 Objectives

To build a scalable, real-time data pipeline and machine learning system that delivers actionable clinical intelligence across hospital operations.

01 Near Real-Time Processing
Ingest and process patient vital data with sub-second pipeline latency using Kafka + PySpark streaming.

02 High-Risk Detection
Apply ML-based deterioration models to score patients continuously and flag high-risk cases.

03 Actionable Dashboards
Surface KPIs and clinical metrics via live dashboards for doctors, nurses, and ICU management.

04 Automated Alerting
Trigger real-time alerts to medical teams when a patient's risk score crosses defined thresholds.

05 Scalable Architecture
Design a cloud-native pipeline capable of handling thousands of concurrent patient data streams.

3 Pre-requisites

The following technical competencies are expected from team members involved in building, maintaining, or extending this platform:

- **Python & SQL:** Writing transformation logic, data validation rules, and ad-hoc queries across pipeline layers.
- **ETL Concepts:** Understanding of Extract, Transform, Load workflows and data pipeline orchestration patterns.
- **PySpark DataFrames:** Distributed data processing for large-scale streaming and batch workloads on Databricks.
- **Cloud Storage:** Working with Azure Data Lake Storage (ADLS) for staging, Bronze, Silver, and Gold layer data.

- **Medallion Architecture:** Familiarity with the Bronze / Silver / Gold layer pattern for data lakehouse design.
- **Streaming Concepts:** Basic knowledge of Kafka or equivalent message brokers for real-time data ingestion.

4 Source Datasets

Five source datasets feed into the pipeline, each capturing a distinct dimension of patient and hospital activity. All streams are ingested in real time into the Bronze layer.

Dataset	Layer	Key Fields
Vitals Stream	Bronze → Silver → Gold	`patient_id · timestamp · heart_rate · spo2 · blood_pressure · temperature`
Patient Registry	Silver (Join)	`patient_id · name · age · ward · admission_date · primary_condition`
Alert Events	Gold	`alert_id · patient_id · alert_time · alert_type · severity · resolved`
ICU Capacity	Gold	`unit_id · total_beds · occupied_beds · last_updated · floor`
Interventions	Gold	`intervention_id · patient_id · action · clinician_id · response_time_min`

5 Architecture — Medallion Pattern

The platform is built on the Medallion Architecture pattern — an industry-standard approach for organizing data lakehouse layers by quality and purpose. Each layer serves a distinct role in the data lifecycle from raw ingestion to business-ready insights.

BRONZE LAYER — Raw Data Ingestion

The Bronze layer stores raw patient vitals exactly as received from Kafka streams and IoT bedside sensors. No transformations are applied — data is preserved in its original form inside Delta Lake tables. This guarantees full auditability and allows the pipeline to be reprocessed if downstream logic changes. Schema is auto-inferred and all records are stamped with an ingestion timestamp.

- All raw data lands here first — no cleaning, no filtering.
- Delta Lake provides ACID transactions, preventing partial writes or data corruption.
- Supports full historical replay if Silver or Gold logic is updated.

SILVER LAYER — Cleaned & Standardized Data

The Silver layer applies data quality rules to the raw Bronze data: deduplication of repeated sensor readings, handling of null or physiologically impossible vital values, standardization of timestamp formats, and unit normalization (e.g., blood pressure in mmHg). Patient registry data is joined here to enrich vital streams with patient context — ward, age, primary diagnosis.

- Removes duplicate records and out-of-range sensor noise.
- Joins patient demographics and ward assignment for enriched context.
- Produces a clean, queryable dataset consumed by both Gold and ML layers.

GOLD LAYER — Business Insights & ML Scoring

The Gold layer aggregates cleaned Silver data into clinical KPIs and runs machine learning models for risk scoring. Outputs include rolling vital averages, deterioration risk scores per patient, alert frequency trends, and ICU utilization metrics. This layer is the source of truth for all dashboards, automated clinical alerts, and reporting consumed by medical staff and hospital administrators.

- Aggregated metrics: rolling averages, peak load windows, utilization rates.
- ML risk scores updated continuously as new vitals arrive from the Silver layer.
- Feeds dashboards, alerting systems, and downstream reporting exports.

6 Expected Reports (Gold Layer)

The following five reports are generated from the Gold layer and are designed for consumption by clinical staff, ICU management, and hospital operations teams.

#	Report Name	Description
01	**Hourly Patient Vitals Summary**	Aggregated averages of heart rate, SpO2, blood pressure, and temperature per patient per hour. Enables trend analysis and historical review by clinicians.
02	**Deterioration Risk Score Report**	ML-generated risk scores updated in real time for each patient. Flags patients crossing critical risk thresholds for immediate clinical review.
03	**Alert Response Time Report**	Measures the time between an alert being raised and clinical intervention being logged. Used to audit and improve response workflows.
04	**ICU Capacity Utilization Report**	Tracks occupied vs. available ICU beds per floor and unit. Helps operations teams make proactive admission and discharge decisions.
05	**High-Risk Patient Detection Report**	Daily summary of all patients flagged as high-risk by the ML model, with supporting vital evidence and recommended action flags.

7 End-to-End Data Flow

The diagram below illustrates the complete journey of patient data from source systems through the Medallion layers to end consumers.

SOURCE SYSTEMS

Bedside monitors · Wearables · EHR systems · Nurse stations

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LANDING ZONE

Apache Kafka + Azure Data Lake Storage (ADLS) · Real-time event streaming

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BRONZE LAYER

Raw Delta Tables · No transformations · Full audit trail

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SILVER LAYER

Cleaned · Validated · Deduplicated · Patient-enriched data

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GOLD LAYER

Aggregated metrics · ML risk scores · Clinical KPIs · Reports

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END CONSUMERS

Clinicians · ICU Teams · Hospital Operations · Administrators

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This document is prepared for client delivery and review. All architecture decisions, dataset schemas, and report specifications are subject to refinement based on client feedback and data availability.

Data Engineering Team
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